

Лабораторная работа №3

Агентное моделирование

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Содержание

1	Цель работы	5
2	Задание	6
3	Выполнение лабораторной работы	7
4	Выводы	31

Список иллюстраций

3.1	Инициализация проекта	7
3.2	Установка пакетов Agents	7
3.3	Загрузка доп. пакетов	8
3.4	Создание файла для базовых определений	8
3.5	Создание исполняемого файла	8
3.6	Тепловая карта - базовая визуализация	9
3.7	Код из jupyter notebook	9
3.8	Создание и запуск анимации модели	10
3.9	Результат	10
3.10	Файл с динамикой числа маргариток	10
3.11	График динамики	11
3.12	Код из jupyter notebook	11
3.13	Создание файла с комплексным графиком	12
3.14	Комплексный график изменения числа маргариток	12
3.15	Код из jupyter notebook	13
3.16	Базовая визуализация с параметрами	13
3.17	Код из jupyter notebook и еще одна тройка графиков	14
3.18	Динамика изменения числа маргариток с параметрами	14
3.19	Код из jupyter notebook и графики динамики	15
3.20	Файлы с комплексными графиками с параметрами	15
3.21	Код из jupyter notebook и комплексные графики	16

Список таблиц

1 Цель работы

Изучить в теории и на практике возможности агентного моделирования.

2 Задание

Применить агентное моделирование на модели daisyworld.

3 Выполнение лабораторной работы

Создаем рабочий каталог и инициализируем проект в julia (рис. 3.1).

```
julia> using Pkg

julia> Pkg.add("DrWatson")
Updating registry at "C:\Users\Andrew\.julia\registries\General.toml"
Resolving package versions...
Installed Xorg.Libcippiaccess.jll - v0.18.1+0
Installed Libwa.jll - v2.23.0+0
Installed Libdwa.jll - v2.4.125+1
Project No packages added to or removed from "C:\Users\Andrew\.julia\environments\v1.12\Project.toml"
Updating "C:\Users\Andrew\.julia\environments\v1.12\Manifest.toml"
[5d5dc0d] + Xorg.Libcippiaccess.jll v0.18.1+0
[8e33e028] + Libdwa.jll v2.4.125+1
[9a156e7d] + Libwa.jll v2.23.0+0
Precompiling packages finished.
3 dependencies successfully precompiled in 30 seconds. 360 already precompiled.

julia> using DrWatson

julia> initialize_project("project"; authors="Dane Hwa", git=false)
```

Рисунок 3.1: Инициализация проекта

Устанавливаем все необходимые пакеты (Agents) (рис. 3.2).

```
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab83\project> julia

Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.
Version 1.12.0 (2025-10-07)
Official https://julialang.org release

(@v1.12) pkg> add Agents
Resolving package versions...
Installed RealDot - v0.1.0
Installed OnlineStatsBase - v1.7.1
Installed LightSumTypes - v5.2.2
Cloning [94ee1d12-ae83-5a48-8b1c-48b8ff168ae0] Quaternions from https://github.com/JuliaGeometry/Quaternions.jl.git
Installed Quaternions - v0.7.7
Cloning [86aed3b0-92cc-5e75-8d2d-829cb8a9cfe8] DataStructures from https://github.com/JuliaCollections/DataStructures.jl.git
Installed DataStructures - v0.18.22
Cloning [86223c79-3864-5bf0-83f7-82e725a168b6] Graphs from https://github.com/JuliaGraphs/Graphs.jl.git
Installed Graphs - v1.13.1
Cloning [66ada85e-f475-11e3-91d0-f70cc89e6671] Agents from https://github.com/JuliaDynamics/Agents.jl.git
Installed Agents - v7.0.0
Cloning [92933f4c-e287-5a95-a399-4b506db050ca] ProgressMeter from https://github.com/timholly/ProgressMeter.jl.git
Installed ProgressMeter - v1.11.0
Cloning [ff63dad9-3335-55d8-95ec-f8139d39e460] StreamSampling from https://github.com/JuliaDynamics/StreamSampling.jl.git
Installed StreamSampling - v0.7.6
Cloning [6938ab19-8711-5258-84ad-4b1128ba62dc] Rotations from https://github.com/JuliaGeometry/Rotations.jl.git
Installed Rotations - v1.7.1
```

Рисунок 3.2: Установка пакетов Agents

Устанавливаем дополнительные пакеты для работы с агентным моделированием (рис. 3.3).

```
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia
Documentation: https://docs.julialang.org
Type "?" for help, "*" for Pkg help.
Version 1.12.0 (2025-10-07)
Official https://julialang.org release

julia> pkg
ERROR: UndefinedError: 'pkg' not defined in 'Main'
Suggestion: check for spelling errors or missing imports.

(Pv1.12) pkg> add StatsBase
Resolving package versions...
Updating C:\Users\Andrew\julia\environments\v1.12\Project.toml
[2913b0d2] + StatsBase v0.34.18
Manifest No packages added to or removed from 'C:\Users\Andrew\julia\environments\v1.12\Manifest.toml'

(Pv1.12) pkg> *C
julia> exit()
```

Рисунок 3.3: Загрузка доп. пакетов

Создаем файл в папке `src`, в котором мы определим тип агента и функции шага модели (рис. 3.4).

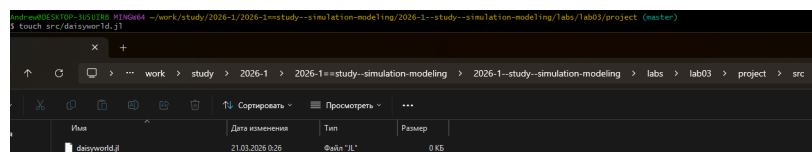


Рисунок 3.4: Создание файла для базовых определений

Создаем исполняемый файл в папке `scripts` и прописываем в нем код (рис. 3.5).

```
Andrew@DESKTOP-3U5UIR8 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/l
$ touch scripts/daisyworld.jl

daisyworld.jl
Файл Изменить Просмотр
using DataFrames
using Plots

include(scrdir("daisyworld.jl"))

using CairoMakie
model = daisyworld()

daisycolor(a::Daisy) = a.breed

plotkwargs = (
    agent_color=daisycolor, agent_size = 20, agent_marker = '☆',
    heatarray = :temperature,
    heatkwargs = (colorrange = (-20, 60)),,
)
plt1, _ = abmplot(model; plotkwargs...)

step!(model, 5)
plt2, _ = abmplot(model; heatarray = model.temperature, plotkwargs...)

step!(model, 40)
plt3, _ = abmplot(model; heatarray = model.temperature, plotkwargs...)

save(plotsdir("daisy_step001.png"), plt1)
save(plotsdir("daisy_step005.png"), plt2)
save(plotsdir("daisy_step040.png"), plt3)
```

Рисунок 3.5: Создание исполняемого файла

Запускаем файл и анализируем тепловую карту, которая будет представлять собой температуру поверхности модели. Маргаритки будут отображаться черно-белыми в

соответствии с их видом. (рис. 3.6).

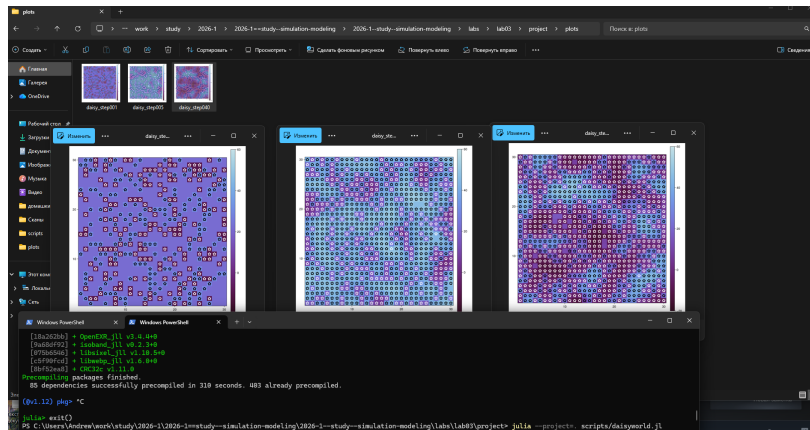


Рисунок 3.6: Тепловая карта - базовая визуализация

Преобразовываем код в произвольные стили, открываем jupyter файл и убеждаемся в правильной компиляции. (рис. 3.7).

```
PS C:\Users\Andrey\work\study\2026-1\study-simulation-modeling\2026-1-study-simulation-modeling\lab03\project> julia --project= scripts\style.jl scripts\daisyworld.jl
Info: generating plain script file from "C:\Users\Andrey\work\study\2026-1\study-simulation-modeling\2026-1-study-simulation-modeling\lab03\project\scripts\daisyworld.jl"
Info: writing result to "C:\Users\Andrey\work\study\2026-1\study-simulation-modeling\2026-1-study-simulation-modeling\lab03\project\scripts\daisyworld\daisyworld.jl"
Info: generating notebook from "C:\Users\Andrey\work\study\2026-1\study-simulation-modeling\2026-1-study-simulation-modeling\lab03\project\scripts\daisyworld.jl"
Info: writing result to "C:\Users\Andrey\work\study\2026-1\study-simulation-modeling\2026-1-study-simulation-modeling\lab03\project\notebooks\daisyworld\daisyworld.qmd"
Info: generating notebook from "C:\Users\Andrey\work\study\2026-1\study-simulation-modeling\2026-1-study-simulation-modeling\lab03\project\scripts\daisyworld.jl"
Info: writing result to "C:\Users\Andrey\work\study\2026-1\study-simulation-modeling\2026-1-study-simulation-modeling\lab03\project\notebooks\daisyworld\daisyworld.ipynb"
Info: Notebook: "C:\Users\Andrey\work\study\2026-1\study-simulation-modeling\2026-1-study-simulation-modeling\lab03\project\notebooks\daisyworld\daisyworld.ipynb"
Format! See help on @show.
PS C:\Users\Andrey\work\study\2026-1\study-simulation-modeling\2026-1-study-simulation-modeling\lab03\project>

plotkwargs = {
    agent_color=:discolor, agent_size = 20, agent_marker = "x",
    heatmap = :temperature,
    heatmap_colorrange = (-20, 40),
}
plt1, _ = simulate(model; plotkwargs...)

step(model, 1)
plt2, _ = simulate(model; heatmap = model.temperature, plotkwargs...)

step(model, 40)
plt3, _ = simulate(model; heatmap = model.temperature, plotkwargs...)

savefigstyle("daisy_step001.png", plt1)
savefigstyle("daisy_step40.png", plt2)
savefigstyle("daisy_step40.png", plt3)
```

Рисунок 3.7: Код из jupyter notebook

Создаем файл с анимацией, прописываем в нем код и запускаем. (рис. 3.8).

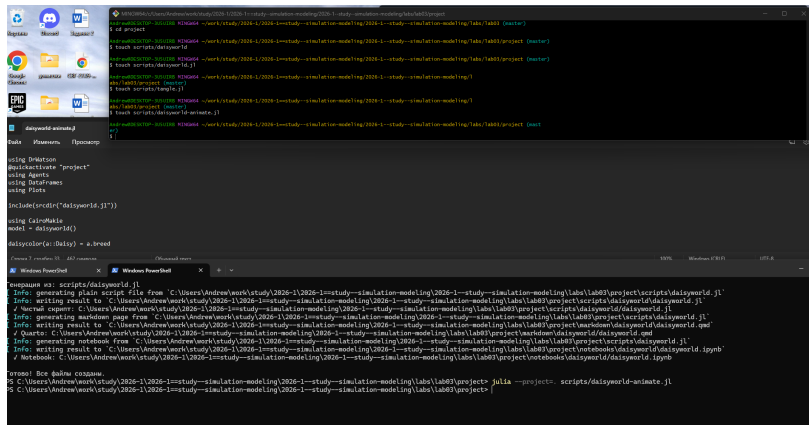


Рисунок 3.8: Создание и запуск анимации модели

Результат можно увидеть в папке plots или по ссылке в rutube: <https://rutube.ru/video/fc59731ceb08176e1f6a8170228c0836/> (рис. 3.9).

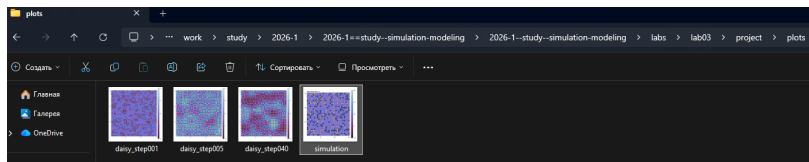


Рисунок 3.9: Результат

<https://rutube.ru/video/fc59731ceb08176e1f6a8170228c0836/>

Создаем, заполняем и запускаем файл, в котором наглядно показана динамика числа маргариток (рис. 3.10).

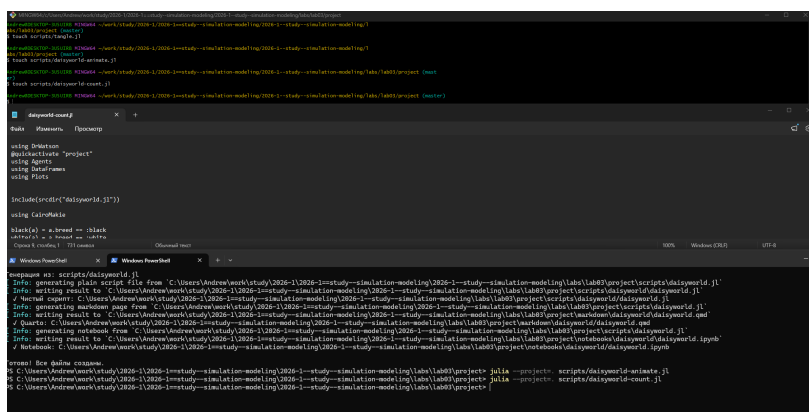


Рисунок 3.10: Файл с динамикой числа маргариток

Наглядно видим результат запуска файла. (рис. 3.11).

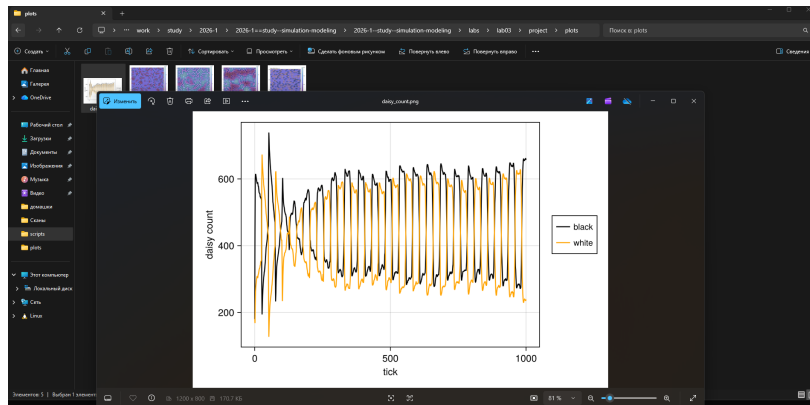


Рисунок 3.11: График динамики

Преобразовываем код в произвольные стили, открываем jupyter файл и убеждаемся в правильной компиляции. (рис. 3.12).

```
from pathlib import Path
import os
import sys
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

# Define the path to the data directory
data_dir = Path('data')
# Create the data directory if it doesn't exist
data_dir.mkdir(exist_ok=True)

# Define the path to the output directory
output_dir = Path('output')
# Create the output directory if it doesn't exist
output_dir.mkdir(exist_ok=True)

# Define the path to the input file
input_file = Path('input.csv')

# Load the input data
input_data = pd.read_csv(input_file)

# Define the path to the output file
output_file = Path('output.csv')

# Define the path to the log file
log_file = Path('log.txt')

# Define the path to the model file
model_file = Path('model.py')

# Define the path to the script file
script_file = Path('script.py')

# Define the path to the notebook file
notebook_file = Path('notebook.ipynb')

# Define the path to the project file
project_file = Path('project.py')

# Define the path to the environment file
environment_file = Path('environment.yml')

# Define the path to the requirements file
requirements_file = Path('requirements.txt')

# Define the path to the README file
readme_file = Path('README.md')

# Define the path to the LICENSE file
license_file = Path('LICENSE.txt')

# Define the path to the .gitignore file
gitignore_file = Path('.gitignore')
```

Рисунок 3.12: Код из jupyter notebook

Создаем, заполняем и запускаем файл, который построит комплексный график изменения числа маргариток, температуры, альбедо в зависимости от модельного времени (рис. 3.13).

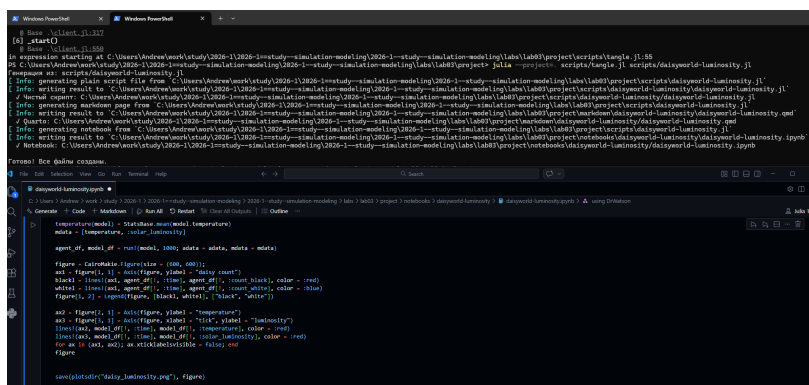


Рисунок 3.15: Код из jupyter notebook

Мы можем расширить базовую визуализацию за счет параметров. Для этого создаем новый файл, в котором прописываем параметры и на выходе получаем несколько троек графиков для каждого набора параметров (рис. 3.16).

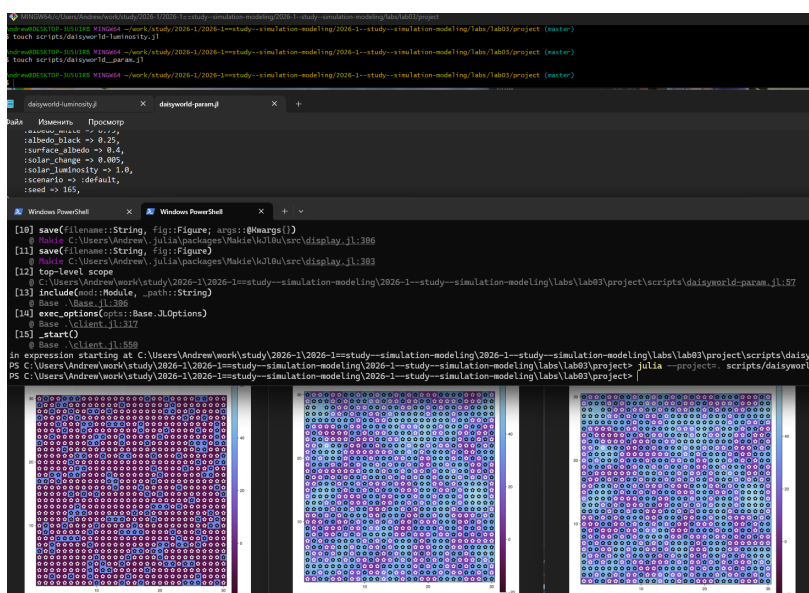


Рисунок 3.16: Базовая визуализация с параметрами

Преобразовываем код в произвольные стили, открываем jupyter файл и убеждаемся в правильной компиляции. (рис. 3.17).

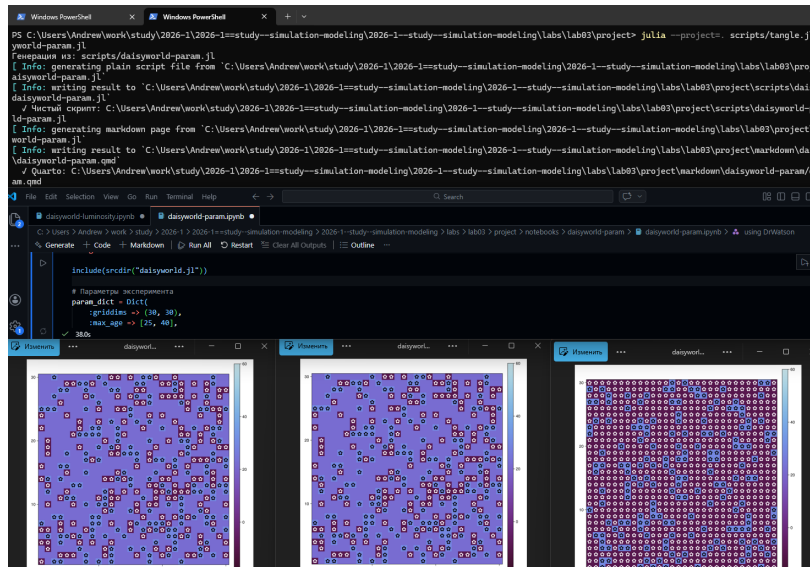


Рисунок 3.17: Код из jupyter notebook и еще одна тройка графиков

Создаем файл, в котором зависимо от заданного набора параметров можно построить график динамика числа маргаритов в зависимости от времени, заполняем и запускаем его (рис. 3.18).

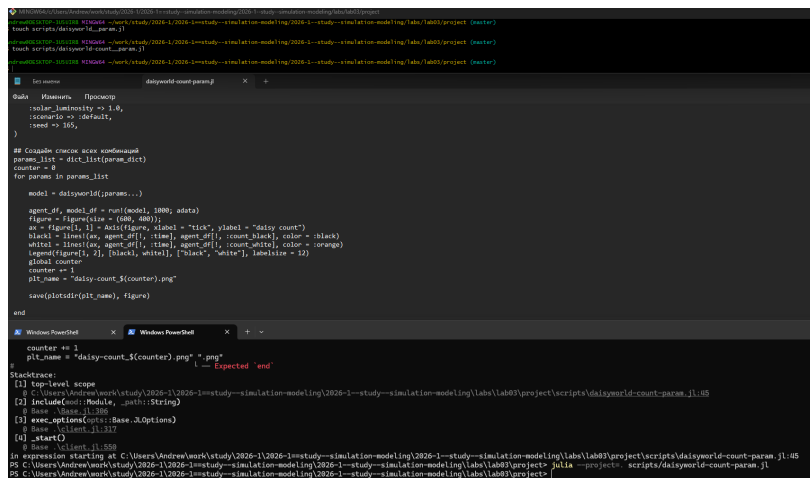


Рисунок 3.18: Динамика изменения числа маргариток с параметрами

Преобразовываем код в произвольные стили, открываем jupyter файл. анализируем получившиеся графики, зависящие от наших заданных параметров (рис. 3.19).

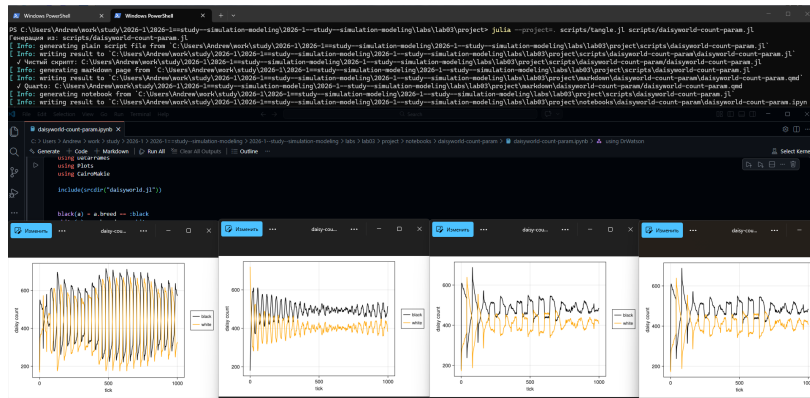


Рисунок 3.19: Код из jupyter notebook и графики динамики

Создаем, заполняем и запускаем файл, в котором можно построить комплексные графики, задав произвольные параметры (рис. 3.20).

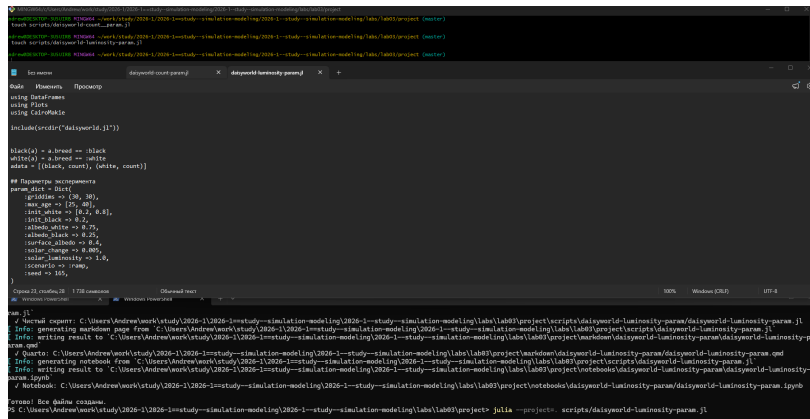


Рисунок 3.20: Файлы с комплексными графиками с параметрами

Преобразовываем код в произвольные стили, открываем jupyter файл. Анализируем получившиеся графики,зависящие от наших заданных параметров (рис. 3.21).

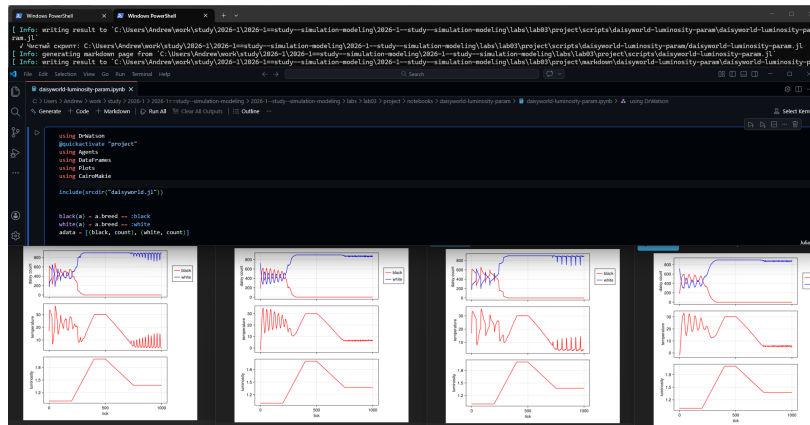


Рисунок 3.21: Код из jupyter notebook и комплексные графики

```
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots

include(sourcedir("daisyworld.jl"))

using CairoMakie

#XXXXXXXXXX XXXXXXXX

model = daisyworld()

daisycolor(a::Daisy) = a.breed

plotkwargs = (
    agent_color=daisycolor, agent_size = 20, agent_marker = 'X',
    heatarray = :temperature,
    heatkwargs = (colorrange = (-20, 60),),
```



```

)

#XXXXXXXXXX XXXXXXXXX

plt1, _ = abmplot(model; plotkwargs...)

step!(model, 5)
plt2, _ = abmplot(model; heatarray = model.temperature, plotkwargs...)

step!(model, 40)
plt3, _ = abmplot(model; heatarray = model.temperature, plotkwargs...)

#XXXXXXXXXXXX XXXXXXXXXXXX

save(plotsdir("daisy_step001.png"), plt1)
save(plotsdir("daisy_step005.png"), plt2)
save(plotsdir("daisy_step040.png"), plt3)

```

```

└ Warning: DrWatson could not find a project file by recursively cl
⊠ (given dir: C:\Users\AESoldatov)
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\
└ Warning: Using the standard Julia project.
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\
└ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\AESoldatov\.julia\packages\Makie\kJl0u\src\scenes
└ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\AESoldatov\.julia\packages\Makie\kJl0u\src\scenes
└ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\AESoldatov\.julia\packages\Makie\kJl0u\src\scenes
└ Warning: Using the standard Julia project.

```

```
└─ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\
└─ Warning: Using the standard Julia project.
└─ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\
└─ Warning: Using the standard Julia project.
└─ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\
```

```
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots

include(sourcedir("daisyworld.jl"))

using CairoMakie

#XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]

#XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
model = daisyworld(; solar_luminosity = 1.0)

#XXXXXXX XXXXXXXXXXXXXXXX X XXXXX XXXXXXXX
agent_df, model_df = run!(model, 1000; adata)

#XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
figure = Figure(size = (600, 400));
```

```

ax = figure[1, 1] = Axis{Figure}(figure, xlabel = "tick", ylabel = "daisy c
blackl = lines!(ax, agent_df[:, :time], agent_df[:, :count_black], c
whitel = lines!(ax, agent_df[:, :time], agent_df[:, :count_white], c
Legend{Figure}(figure[1, 2], [blackl, whitel], ["black", "white"], labelsiz

#XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
save(plotsdir("daisy_count.png"), figure)

```

```

└─ Warning: DrWatson could not find a project file by recursively cl
⊠ (given dir: C:\Users\AESoldatov)
└─ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\j
└─ Warning: Using the standard Julia project.
└─ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\j
└─ Warning: Found `resolution` in the theme when creating a `Scene`. The
└─ @ Makie C:\Users\AESoldatov\.julia\packages\Makie\kJl0u\src\scenes
└─ Warning: Using the standard Julia project.
└─ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\j

```

```

using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots

include(scrdir("daisyworld.jl"))

using CairoMakie

```

```

# 定义初始数据
black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]

# 定义模型
model = daisyworld(solar_luminosity = 1.0, scenario = :ramp)

# 定义统计量
temperature(model) = StatsBase.mean(model.temperature)
mdata = [temperature, :solar_luminosity]

# 运行模型
agent_df, model_df = run!(model, 1000; adata = adata, mdata = mdata)

# 绘图
figure = CairoMakie.Figure(size = (600, 600));
ax1 = figure[1, 1] = Axis(figure, ylabel = "daisy count")
blackl = lines!(ax1, agent_df[:, :time], agent_df[:, :count_black],
whitel = lines!(ax1, agent_df[:, :time], agent_df[:, :count_white],
figure[1, 2] = Legend(figure, [blackl, whitel], ["black", "white"])

ax2 = figure[2, 1] = Axis(figure, ylabel = "temperature")
ax3 = figure[3, 1] = Axis(figure, xlabel = "tick", ylabel = "luminosity")

lines!(ax2, model_df[:, :time], model_df[:, :temperature], color = :blue)
lines!(ax3, model_df[:, :time], model_df[:, :solar_luminosity], color = :red)
for ax in (ax1, ax2); ax.xticklabelsvisible = false; end

```

```
figure
```

```
#XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
save(plotsdir("daisy_luminosity.png"), figure)
```

```
└ Warning: DrWatson could not find find a project file by recursively cl  
⊠ (given dir: C:\Users\AESoldatov)
```

```
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\j
```

```
└ Warning: Using the standard Julia project.
```

```
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\j
```

```
└ Warning: Found `resolution` in the theme when creating a `Scene`. The
```

```
└ @ Makie C:\Users\AESoldatov\.julia\packages\Makie\kJl0u\src\scenes
```

```
└ Warning: Using the standard Julia project.
```

```
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\j
```

```
using DrWatson
```

```
@quickactivate "project"
```

```
using Agents
```

```
using DataFrames
```

```
using Plots
```

```
using CairoMakie
```

```
include(sourcedir("daisyworld.jl"))
```

```
# XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
param_dict = Dict(  
    :griddims => (30, 30),  
    :max_age => [25, 40],  
    :init_white => [0.2, 0.8],
```

```

: init_black => 0.2,
: albedo_white => 0.75,
: albedo_black => 0.25,
: surface_albedo => 0.4,
: solar_change => 0.005,
: solar_luminosity => 1.0,
: scenario => :default,
: seed => 165,
)

# 1000000 1000000 10000 10000000000

params_list = dict_list(param_dict)

using SHA

for params in params_list
    model = daisyworld(;params...)

    daisycolor(a::Daisy) = a.breed
    plotkwargs = (
        agent_color=daisycolor,
        agent_size = 20,
        agent_marker = '█',
        heatarray = :temperature,
        heatkwargs = (colorrange = (-20, 60),),
    )
    plt1, _ = abmplot(model; plotkwargs...)

```

```

    step!(model, 5)
    plt2, _ = abmplot(model; heatarray = model.temperature, plotkwa

    step!(model, 40)
    plt3, _ = abmplot(model; heatarray = model.temperature, plotkwa

    hash_part = bytes2hex(sha256(string(params)))[1:8]

    plt1_name = "daisyworld_$(hash_part)_step01.png"
    plt2_name = "daisyworld_$(hash_part)_step04.png"
    plt3_name = "daisyworld_$(hash_part)_step40.png"

    save(plotsdir(plt1_name), plt1)
    save(plotsdir(plt2_name), plt2)
    save(plotsdir(plt3_name), plt3)

end

```

```

└ Warning: DrWatson could not find find a project file by recursively cl
⊠ (given dir: C:\Users\AESoldatov)
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\j
└ Warning: Using the standard Julia project.
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└ @ Makie C:\Users\AESoldatov\.julia\packages\Makie\kJl0u\src\scenes
└ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\AESoldatov\.julia\packages\Makie\kJl0u\src\scenes

```

```

└ Warning: Using the standard Julia project.
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\
└ Warning: Using the standard Julia project.
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└ Warning: Found `resolution` in the theme when creating a `Scene`. The
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└ Warning: Using the standard Julia project.
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\
└ Warning: Using the standard Julia project.

```



```

└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\
└ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\AESoldatov\.julia\packages\Makie\kJl0u\src\scenes
└ Warning: Found `resolution` in the theme when creating a `Scene`. The
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└ Warning: Using the standard Julia project.
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\

```

```

using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
using CairoMakie

include(scrdir("daisyworld.jl"))

black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]

# ██████████ ████████████████████

param_dict = Dict(

```

```

:griddims => (30, 30),
:max_age => [25, 40],
:init_white => [0.2, 0.8],
:init_black => 0.2,
:albedo_white => 0.75,
:albedo_black => 0.25,
:surface_albedo => 0.4,
:solar_change => 0.005,
:solar_luminosity => 1.0,
:scenario => :default,
:seed => 165,
)

# 测试参数 测试值 测试 测试值
params_list = dict_list(param_dict)

using SHA

for params in params_list
    model = daisyworld(;params...)
    agent_df, model_df = run!(model, 1000; adata)
    figure = Figure(size = (600, 400));
    ax = figure[1, 1] = Axis(figure, xlabel = "tick", ylabel = "daisy")
    blackl = lines!(ax, agent_df[:, :time], agent_df[:, :count_black])
    whitel = lines!(ax, agent_df[:, :time], agent_df[:, :count_white])
    Legend(figure[1, 2], [blackl, whitel], ["black", "white"], label = "count")

    hash_part = bytes2hex(sha256(string(params)))[1:8]

```

```

plt_name = "daisy-count_$(hash_part).png"

save(plotsdir(plt_name), figure)

end

```

```

└ Warning: DrWatson could not find find a project file by recursively cl
⊠ (given dir: C:\Users\AESoldatov)
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\j
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└ Warning: Using the standard Julia project.
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\j

```

```

using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
using CairoMakie

include(sourcedir("daisyworld.jl"))

black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]

# 地形データを読み込む
param_dict = Dict(
    :griddims => (30, 30),
    :max_age => [25, 40],
    :init_white => [0.2, 0.8],
    :init_black => 0.2,
    :albedo_white => 0.75,
    :albedo_black => 0.25,
    :surface_albedo => 0.4,
    :solar_change => 0.005,
    :solar_luminosity => 1.0,
    :scenario => :ramp,
    :seed => 165,
)

```

```

# 测试 模型 参数 列表
params_list = dict_list(param_dict)

using SHA

for params in params_list
    model = daisyworld(;params...)

    temperature(model) = StatsBase.mean(model.temperature)
    mdata = [temperature, :solar_luminosity]

    agent_df, model_df = run!(model, 1000; adata = adata, mdata = mdata)

    figure = CairoMakie.Figure(size = (600, 600));
    ax1 = figure[1, 1] = Axis(figure, ylabel = "daisy count")
    blackl = lines!(ax1, agent_df[:, :time], agent_df[:, :count_black], color = "black")
    whitel = lines!(ax1, agent_df[:, :time], agent_df[:, :count_white], color = "white")
    figure[1, 2] = Legend(figure, [blackl, whitel], ["black", "white"])

    ax2 = figure[2, 1] = Axis(figure, ylabel = "temperature")
    ax3 = figure[3, 1] = Axis(figure, xlabel = "tick", ylabel = "luminosity")
    lines!(ax2, model_df[:, :time], model_df[:, :temperature], color = "red")
    lines!(ax3, model_df[:, :time], model_df[:, :solar_luminosity], color = "blue")
    for ax in (ax1, ax2); ax.xticklabelsvisible = false; end

    hash_part = bytes2hex(sha256(string(params)))[1:8]

    plt_name = "daisy-luminosity_$(hash_part).png"
end

```

```
save(plotsdir(plt_name), figure)
```

```
end
```

```
└ Warning: DrWatson could not find find a project file by recursively cl
⊠ (given dir: C:\Users\AESoldatov)
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\j
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└ Warning: Using the standard Julia project.
└ @ DrWatson C:\Users\AESoldatov\.julia\packages\DrWatson\2QF5p\src\j
```

4 Выводы

В данной работе мы разобрались с агентным моделированием и реализовали его на практике.